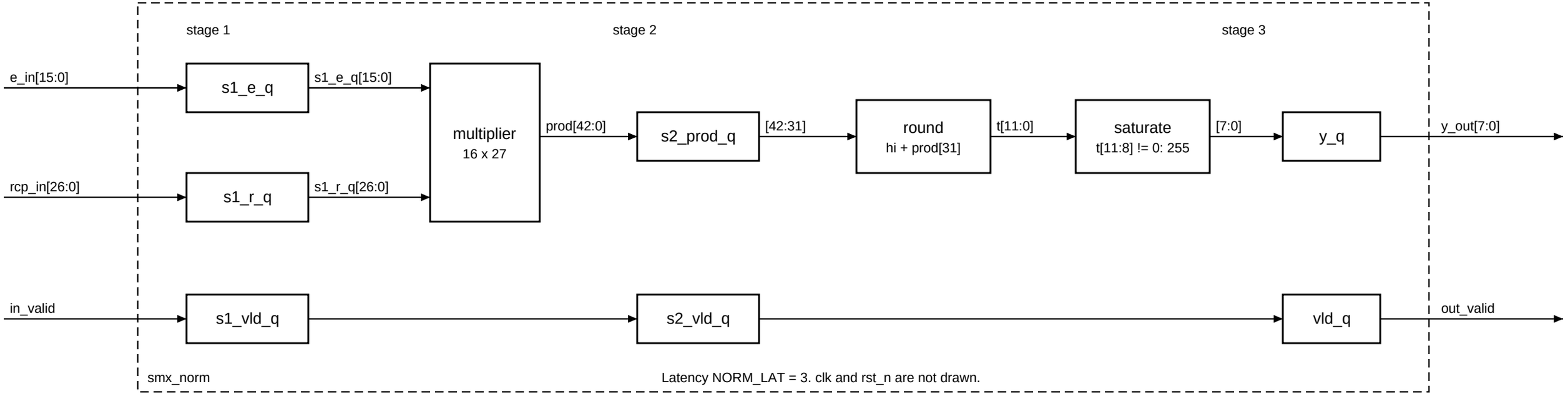
stage 1
stage 2
stage 3
e_in[15:0]
s1_e_q
s1_e_q[15:0]
multiplier
16 x 27
prod[42:0]
s2_prod_q
[42:31]
round
hi + prod[31]
t[11:0]
saturate
t[11:8] != 0: 255
[7:0]
y_q
y_out[7:0]
rcp_in[26:0]
s1_r_q
s1_r_q[26:0]
in_valid
s1_vld_q
s2_vld_q
vld_q
out_valid
smx_norm
Latency NORM_LAT = 3. clk and rst_n are not drawn.